

StreamCatTools: An R package for working with StreamCat and LakeCat watershed data in R

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Summary

StreamCatTools provides an R interface to the StreamCat(Hill et al., 2016) and LakeCat(Hill et al., 2018) data and APIs for retrieving national scale watershed and lake-basin metrics in a reproducible workflow. Built on the NHDPlusV21 (McKay et al., 2012) framework, the package gives researchers and managers direct, simple access to hundreds of catchment- and watershed-scale metrics without requiring manual geospatial accumulation.

Statement of Need

Nationally consistent and easily accessible watershed data are essential for hydrology, water-quality assessment, and ecological modeling. The StreamCat (Hill et al., 2016) and LakeCat (Hill et al., 2018) datasets provide such data for every NHDPlusV21 stream reach and lake basin in the conterminous United States (CONUS) and **StreamCatTools** makes these data accessible in R through a simple API wrapper for metric discovery, batched retrieval of data by unique identifier (COMID), state, county, hydroregion, or CONUS. This lowers the barrier to reproducible analysis and supports the FAIR principles (Findable, Accessible, Interoperable, and Reusable) (Wilkinson et al., 2016).

State of the Field

Programmatic access to watershed data in R in recent years had evolved to encompass several facets to *what* a package delivers: the hydrographic fabric (the stream reaches and catchments composing the network itself), observational time series (flow and water quality), general-purpose geospatial layers, and pre-computed landscape characteristics. The first role is well served by **hydrogeofetch** and its network-navigation companion **hydroloom** (Blodgett, 2026), which subset and traverse the NHDPlusV21 network. The second is served by **dataRetrieval** (DeCicco et al., 2026), which retrieves USGS and EPA hydrology and water-quality observations. The third is covered by general fetchers such as **FedData** and **elevatr**, which return raw federal layers (e.g., land cover, elevation, soils) that a user must summarize to catchments themselves.

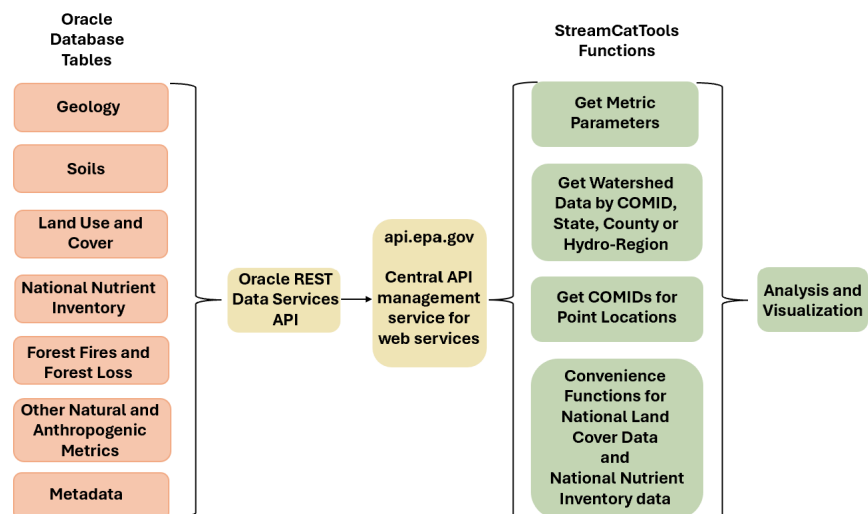


Figure 1: Diagram of the StreamCat and StreamCatTools framework. The backend Oracle database and REST web service are exposed through `api.epa.gov` to functions in StreamCatTools that simplify access and analysis of the data in R via the application programming interface (API). The example here shows the general categories of Oracle database tables in the StreamCat Oracle database.

The remaining role — landscape and anthropogenic metrics *already accumulated* to the local catchment and the full upstream watershed for every NHDPlusV21 or lake reach — has been a gap in the field, yet it is a form most analyses actually need. **StreamCatTools** provides programmatic access to the StreamCat and LakeCat data and API so that users can retrieve analysis-ready watershed covariates in R without performing any flow-network accumulation themselves, and without manually assembling large geospatial layers.

Package Overview

StreamCatTools is a lightweight R interface to the StreamCat and LakeCat APIs. The package exposes functions for metadata discovery, metric lookup, and batched data retrieval from the StreamCat and LakeCat REST services, using `httr2` (Wickham, 2026). The core functions retrieve variables by COMID, state, county, hydroregion, or CONUS, and also support National Land Cover Database (NLCD) (U.S. Geological Survey, 2019) and National Nutrient Inventory (NNI) (Brehob et al., 2025) time-series products.

You can install the most recent version of **StreamCatTools** from *CRAN* by running:

```
install.packages("StreamCatTools")
```

You can install the most recent package version from *GitHub* by running the following code:

```
install.packages("pak")
library(pak)
pkg_install("github::USEPA/StreamCatTools")
```

StreamCatTools is loaded into an **R** session:

```
library(StreamCatTools)
```

Users can inspect available metric names and areas of interest before querying the API. Supported AOIs include catchment (cat), watershed (ws), riparian catchment (catrp100), riparian watershed (wsrp100), and an other category for non-standard metrics. Extracting this information in `StreamCatTools` looks like this:

```
aois <- sc_get_params(param='aoi')
aois
```

```
#> [1] "cat"      "catrp100" "other"     "ws"       "wsrp100"
```

```
names <- sc_get_params(param='metric_names')
names[1:10]
```

```
#> [1] "agkffact"      "al2o3"         "bankfulldepth" "bankfullwidth"
#> [5] "bfi"           "canaldens"     "cao"           "cbnf"
#> [9] "chem"          "clay"
```

The same pattern applies to LakeCat metadata retrieval with `lc_get_params()`.

In StreamCat and LakeCat individual catchments (the local drainage area for a stream or lake) are aggregated to the full upstream watershed using either a weighted average or a count/sum depending on the metric (Hill et al., 2016). `sc_get_params()` and `lc_get_params()` also return variable metadata, including short and long descriptions, units, years, and metric categories (Table 1)

Table 1: Example of variable information returned by the `variable_info` parameter in the `sc_get_params` function.

Metric	Short Description	Year	Units
agkffact[AOI]	Ag Soil Erodibility Kf Factor	NA	Unitless
bfi[AOI]	Base Flow Index	NA	Percent
huden[Year][AOI]	Mean Housing Density	2010	Count/Square Kilometer
inorgnwetdep[Year][AOI]	Mean Annual Precipitation-Weighted Nitrogen	2008	Kilogram/Hectare/Year
n_ags_[Year][AOI]	Nitrogen Agricultural Surplus	1987-2017	Kilograms
pcthighsev[Year][AOI]	Percent High Burn Severity Class For Year	1984-2018	Percent

Additional metadata functions include `sc_fullname()` and `lc_fullname()`, which return the full descriptive name for a given metric.:

```
sc_fullname(metric='pctgrs2019')
```

```
#> [1] "Grassland/Herbaceous Percentage 2019"
```

Users can also filter available metrics by year, category, dataset, and AOI using `sc_get_metric_names()` and `lc_get_metric_names()` (Table 2).

```
metrics <- sc_get_metric_names(category = c('Deposition', 'Climate'),
                              ao=c('Cat', 'Ws'))
my_data <- head(metrics[,c('Category', 'Metric', 'AOI')], 10)
```

Table 2: Example of metric names returned by the `sc_get_metric_names` function.

Category	Metric	AOI
Climate	bfi[AOI]	Cat, Ws
Deposition	inorgnwetdep[Year][AOI]	Cat, Ws
Deposition	nh4[Year][AOI]	Cat, Ws
Deposition	no3[Year][AOI]	Cat, Ws
Climate	precip8110[AOI]	Cat, Ws
Climate	precip9120[AOI]	Cat, Ws
Climate	precip[Year][AOI]	Cat, Ws
Deposition	sn[Year][AOI]	Cat, Ws
Climate	tmax8110[AOI]	Cat, Ws
Climate	tmax9120[AOI]	Cat, Ws

More details on these functions can be found at the [package introduction page](#).

The `sc_get_data()` and `lc_get_data()` functions allow users to extract catchment or watershed metrics by NHDPlusV21 COMID or by spatial filter such as county, state, hydroregion, or CONUS. The following example shows a request for two catchment and watershed metrics for three stream reaches NLCD medium intensity developed land cover and density of dams:

```
df <- sc_get_data(metric='pcturbmd2019,damdens',
                  ao='cat,ws',
                  comid='179,1337,1337420')
```

A parallel LakeCat request can be made by county or other spatial filter, for example by querying county FIPS codes for a state or county-level summary.:

```
df <- lc_get_data(metric='pctwdwet2006', ao='ws', county='41003')
head(df)
```

```
#>      comid pctwdwet2006ws
#> 1 23769033      0.0000000
#> 2 23769029      0.1701645
#> 3 23769021      2.4697951
#> 4 23769003      0.0000000
#> 5 23768999      8.1185071
#> 6 23768983      0.0000000
```

Users can also query by state, county, hydroregion, or CONUS, and helper functions are available to discover the relevant FIPS codes.:

```
df <- sc_get_data(metric='pctwdwet2006', ao='ws', county='41003')
```

The package includes convenience helpers to discover county FIPS codes and to simplify state- and county-level queries.:

```
df <- sc_get_params(param='county') |> dplyr::filter(state == 'OR' & county_name ==
```

Or we can ask for several different metrics for all of a state:

```
# State-wide watershed queries are large; run the live call once, then
# cache to disk so subsequent knits read the cache instead of
# re-hitting (or stalling on) the API. Commit the .rds alongside paper.Rmd.
ct_cache <- "clay_agkffact_CT_ws.rds"
if (file.exists(ct_cache)) {
  df <- readRDS(ct_cache)
} else {
  df <- sc_get_data(metric = 'clay,agkffact', aoi = 'ws', state = 'CT')
  saveRDS(df, ct_cache)
}
head(df)
```

```
#>      comid agkffactws  clayws
#> 1 7702768      0.0177 6.100716
#> 2 7702200      0.0009 6.243493
#> 3 7702202      0.0011 6.368938
#> 4 7701112      0.0177 6.080299
#> 5 7701168      0.0054 6.092434
#> 6 7701214      0.0040 6.082292
```

StreamCatTools also provides convenience functions for the NLCD and NNI datasets through `sc_get_nlcd()`, `lc_get_nlcd()`, `sc_get_nni()`, and `lc_get_nni()`:

```
df <- sc_get_nlcd(comid='1337420', year='2019', aoi='ws')
```

```
df <- sc_get_nni(year='1987, 1990, 2005, 2017', aoi='cat,ws',
  comid='179,1337,1337420')
```

Software design

StreamCatTools is designed as a lightweight wrapper around the StreamCat and LakeCat REST services. Requests are issued through `httr2`, with functions grouped into matching `sc_*` and `lc_*` families for stream and lake metrics. Data access functions accept metric names, spatial filters, and AOI syntax, then return tidy data frames ready for analysis. The package integrates with spatial workflows through `sf` and `hydrogeofetch`, and the lake-basin workflow uses DuckDB-backed geometry access for efficient retrieval.

Applications and Discussion

StreamCatTools is useful for summarizing watershed metrics across many reaches, lakes, or counties and for linking those summaries to spatial analyses and ecological models. For example, users can calculate county- or state-scale summaries of land cover or hydrologic stressors and then combine those outputs with `ggplot2` or `hydrogeofetch`-based mapping such as in the following examples:

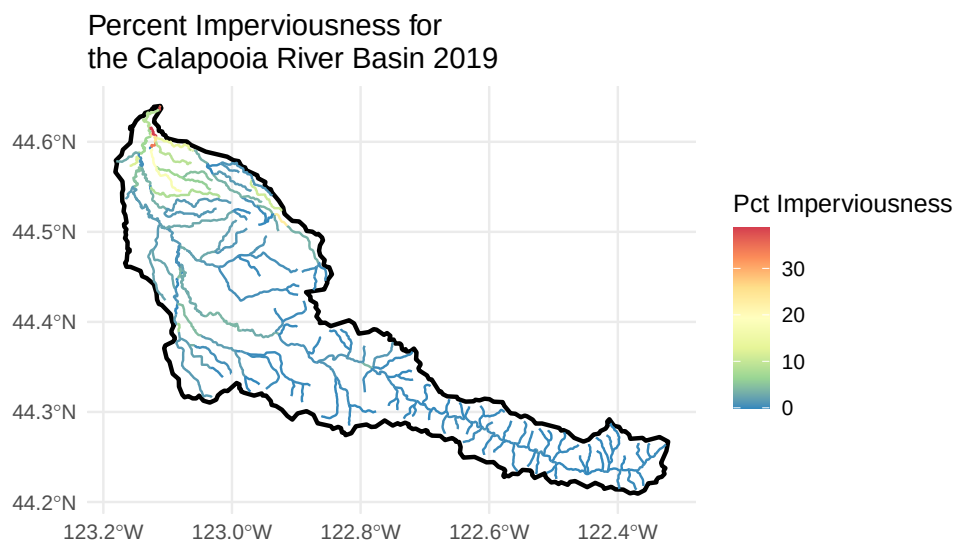


Figure 2: Map of NLCD percent imperviousness for each catchment for the Calapooia River watershed in Oregon.

```
df <- sc_get_data(metric='pctagslphigh2019', aoi='ws', county='41003') |>
  dplyr::summarise(mean_pctagslphigh2019ws = mean(pctagslphigh2019ws, na.rm=TRUE))
df
```

```
#> mean_pctagslphigh2019ws
#> 1 4.406472
```

```
library(hydrogeofetch)
library(ggplot2)
library(ggspatial)
library(StreamCatTools)

start_comid = 23763517
nldi_feature <- list(featureSource = "comid", featureID = start_comid)
flowline_nldi <- hydrogeofetch::navigate_nldi(nldi_feature, mode = "UT", data_source = "nldi")
df <- sc_get_data(metric='pctimp2019', aoi='cat', comid=flowline_nldi$UT_flowlines_comid)
flowline_nldi <- flowline_nldi$UT_flowlines
flowline_nldi$PCTIMP2019 <- df$pctimp2019cat[match(flowline_nldi$nhdplus_comid, df$comid)]
basin <- hydrogeofetch::get_nldi_basin(nldi_feature = nldi_feature)
```

Figure 2 plots the NLCD percent imperviousness (percentage of area covered by constructed, artificial surfaces) for the local drainage (catchment in NHDPlusV21 syntax) mapped to each stream reach and to the overall basin boundary.

Watersheds for lakes can also be retrieved using `lc_get_watershed()` to visualize lake-basin metrics and land-cover composition as shown in Figure 3.

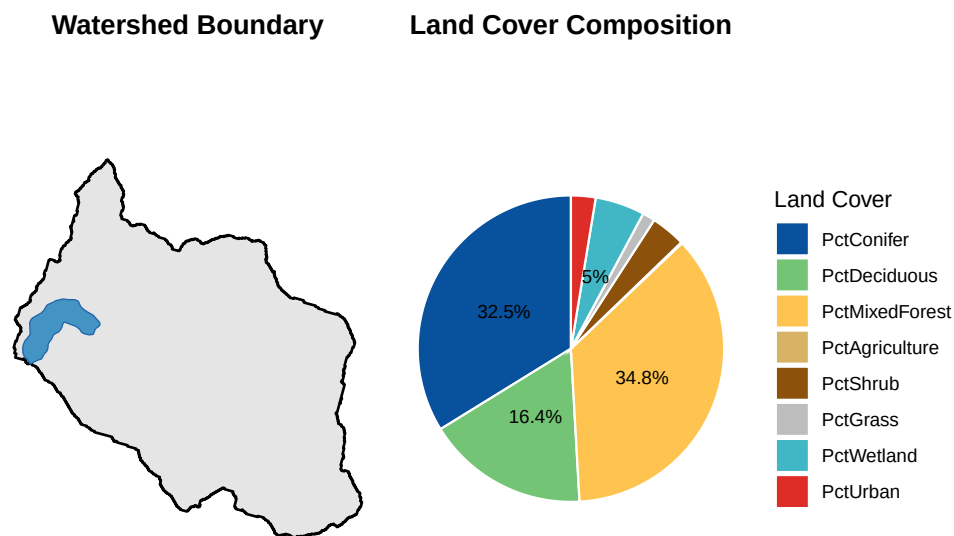


Figure 3: NLCD land cover proportions with an example lake watershed.

```
library(ggplot2)
library(patchwork)
library(ggforce)

df <- lc_get_nlcd(comid='19334077', year='2019', aoi='ws')
lake <- hydrogeofetch::get_waterbodies(id = 19334077)
ws <- lc_get_watershed(comid = 19334077, huc2 = "01", huc2_filter = "01",
                      threads = 2, retries = 5, verbose=FALSE, progress=FALSE)
```

Functions for plotting NNI metrics are also included (Markley et al., 2026), enabling time-series exploration for nitrogen and phosphorus budgets in a selected watershed 4.

```
library(StreamCatTools)
library(ggplot2)
library(ggpattern)
com <- '22812041'
sc_plotnni(comid = com, include.nue = TRUE)
```

Future functionality for **StreamCatTools** includes expanding plotting functions and adding more metrics. Additionally, a complementary package, **StreamCatR** is being developed that to allow users to process their own landscape metrics to the NHDPlus framework or other hydrologic frameworks with network topology. These features support a broader objective of making national-scale watershed data more accessible and reproducible for management and research applications.

StreamCatTools currently facilitates easy ingestion of **StreamCat** and **LakeCat** watershed landscape metrics into workflows in R which is of great use to state watershed planners, researchers and non-governmental organizations and borne out by the over 12000 package downloads and over 350 citations of the underlying **StreamCat** and **LakeCat** data that are served by the **StreamCatTools** package.

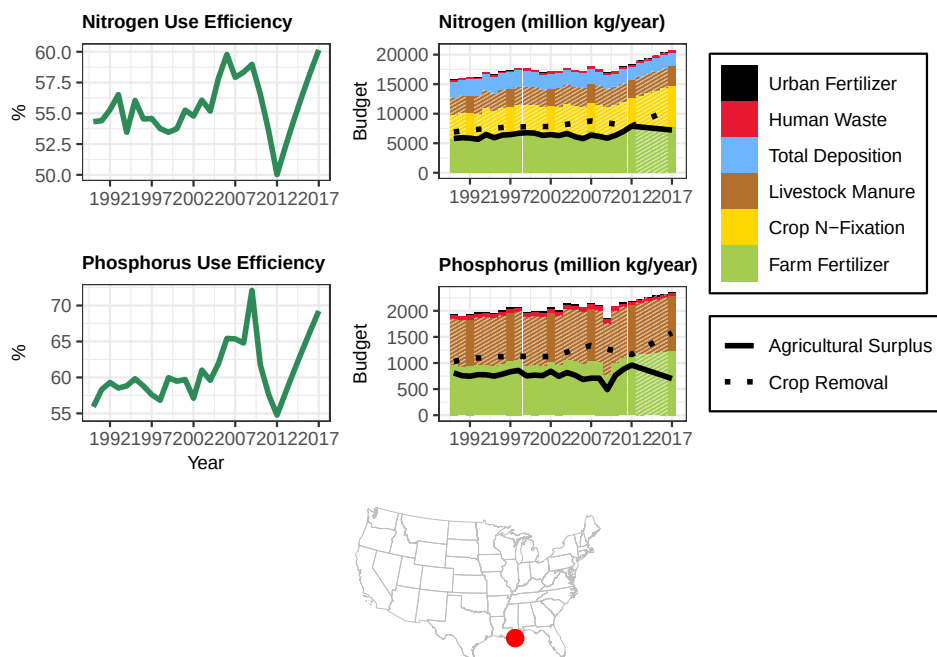


Figure 4: Annual time series of nitrogen and phosphorus budget data for the Mississippi-Atchafalaya River Basin.

Research impact

StreamCatTools and the underlying **StreamCat** dataset provides more than 600 natural and anthropogenic landscape metrics — summarized to both local catchments and full upstream watersheds — for roughly 2.65 million stream reaches across the conterminous United States (Hill et al., 2016), with the parallel **LakeCat** dataset extending the same framework to more than 356,000 lakes. **StreamCatTools** provides reproducible, programmatic watershed summaries from the **StreamCat** and **LakeCat** datasets at the CONUS scale. By providing metric discovery, batched retrieval, and spatial integration in a single R interface, **StreamCatTools** facilitates use of this valuable CONUS scale data in monitoring, ecological assessment, and management decisions. Beyond research, the package and data support management and regulatory applications under Clean Water Act authorities in state integrated reporting (i.e. Oregon Department of Environmental Quality (Oregon Department of Environmental Quality, n.d.) and Alabama Department of Environmental Management (Alabama Department of Environmental Management, 2025)), as well as species-distribution and aquatic-connectivity studies.

Acknowledgements

Examples of using **StreamCat** and **LakeCat** make extensive use of **hydrogeofetch** (Blodgett, 2026) and the functions for accessing the API are facilitated through use of **httr2** (Wickham, 2026). Figures were created using **ggplot2** (Wickham, 2016).

We would like to sincerely thank Michael Dumelle and Jeff Hollister for helpful comments which improved this manuscript as well as the editor and reviewers for all of their helpful feedback which greatly improved both the software and the manuscript.

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AI Usage Disclosure

The R code for this R package was initially developed by Marc Weber and minimal generative AI was used for the most recent version of the package where several functions were refactored.

References

- Alabama Department of Environmental Management. (2025). *Water quality monitoring strategy*. Alabama Department of Environmental Management. <https://adem.alabama.gov/sites/default/files/2025-09/WQMonitoringStrategy.pdf>
- Blodgett, D. (2026). *Hydrogeofetch: Hydrologic geospatial fabric data extraction tool chain* (Version 2.0.0). U.S. Geological Survey; U.S. Geological Survey. <https://github.com/doi-usgs/hydrogeofetch>
- Brehob, M. M., Pennino, M. J., Compton, J. E., Zhang, Q., Weber, M. H., Hill, R. A., Markley, S., Pickard, B., Keefer, M., Stackpoole, S. M., Knose, L. A., Ruiz-Mercado, G. J., Clark, C. M., Rea, A. W., Carleton, J. N., Lin, J., Bash, J. O., Foley, K. M., Hogrefe, C., & Sabo, R. D. (2025). The US EPA’s national nutrient inventory: Critical shifts in US nutrient pollution sources from 1987 to 2017. *Environmental Science & Technology*, 59(51), 27836–27852. <https://doi.org/10.1021/acs.est.5c08196>
- DeCicco, L., Hirsch, R., Lorenz, D., et al. (2026). *dataRetrieval: Retrieval functions for USGS and EPA hydrology and water quality data*. <https://doi-usgs.github.io/dataRetrieval/>
- Hill, R. A., Weber, M. H., Debbout, R. M., Leibowitz, S. G., & Olsen, A. R. (2018). The lake-catchment (LakeCat) dataset: Characterizing landscape features for lake basins within the conterminous USA. *Freshwater Science*, 37(2), 208–221. <https://doi.org/10.1086/697966>
- Hill, R. A., Weber, M. H., Leibowitz, S. G., Olsen, A. R., & Thornbrugh, D. J. (2016). The stream-catchment (StreamCat) dataset: A database of watershed metrics for the conterminous united states. *JAWRA Journal of the American Water Resources Association*, 52(1), 120–128. <https://doi.org/10.1111/1752-1688.12372>
- Markley, S., Weber, M. H., Hill, R. A., Brehob, M., Alford, S., Sabo, R., Pennino, M., & Compton, J. (2026). *StreamCatNNI and LakeCatNNI: Watershed and local catchment nitrogen and phosphorus budgets for US streams and lakes*.
- McKay, L., Bondelid, T., Dewald, T., Johnston, J., Moore, R., & Reah, A. (2012). *NHDPlus version 2: User guide*. http://www.horizon-systems.com/NHDPlus/NHDPlusV2_home.php

- Oregon Department of Environmental Quality. (n.d.). *IR26 TSD*. Oregon Department of Environmental Quality. <https://www.oregon.gov/deq/wq/Documents/ir26tsd.pdf>
- U.S. Geological Survey. (2019). *National land cover database*. <https://www.mrlc.gov/>. <https://doi.org/10.5066/P94UXNTS>
- Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis*. Springer-Verlag New York. <https://doi.org/10.1007/978-0-387-98141-3>
- Wickham, H. (2026). *httr2: Perform HTTP requests and process the responses*. <https://doi.org/10.32614/CRAN.package.httr2>
- Wilkinson, M. D., Dumontier, M., Aalbersberg, Ij. J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.-W., Silva Santos, L. B. da, Bourne, P. E., et al. (2016). The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, 3(1), 1–9. <https://doi.org/10.1038/sdata.2016.18>